

- 26 C** At t_1 , increasing current in coil Q causes increase in B field generated by Q.
The induced emf in coil P is proportional to the rate of increase in magnetic flux at P. By Lenz's Law, induced current in P flows to generate a magnetic field to oppose the increase in magnetic flux.
That results in repulsion of the two coils. Due to the downward magnetic force on coil P, the normal contact force, N will now have the magnitude of (magnetic force + W), greater than W .
- At t_2 and t_4 , $N = mg$ since there is no change in flux, there is no induced emf or current.
- At t_3 , although there is induced current in P, there is no current in Q, thus there is no magnetic interaction, $N = W$.

